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Regionalization of extreme precipitation estimates for the Alabama rainfall atlas

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Abstract

The Alabama Rainfall Atlas is an internet-based server for delivery of design information on the characteristics of extreme precipitation throughout the state. Products delivered include intensity-duration-frequency (IDF) curves and 24-h design storm hyetographs based on US Soil Conservation Service (now the Natural Resources Conservation Service) methods. Three databases (daily, hourly, subhourly) upon which the server relies consist of at-site estimates of generalized extreme value distribution parameters for gauging sites throughout the state. Regionalization of those at-site estimates to develop precipitation products for a user-specified location is accomplished using a hybrid estimation method that consists of a weighted average of estimators based on gauging station proximities and record lengths. The regionalization procedure provides products on demand for any user-selected location (gauged or ungauged) and does not cause discontinuities at boundaries that would result from delineation of distinct regions. The databases are compact and user-requested products are produced and delivered nearly instantaneously. These latter issues are felt to be major ones considering the internet-based delivery method.

Keywords: Precipitation; Frequency analysis; Regionalization; Alabama

1. Introduction

Information on the magnitudes and frequencies of extreme precipitation is essential for design of water conveyance and flood protection structures (storm sewers, culverts, etc.), for agricultural purposes, and for studies related to weather modification and climatic change. For some types of engineering analyses, particularly those related to the rational method of storm runoff estimation, one requires the rainfall intensity-duration-frequency (IDF) relationship applicable to a location of interest. For more complex unit-hydrograph-based analyses the engineer requires information on the time distribution (i.e. the hyetograph) of a design storm of interest. In some cases, particularly where significant storage is involved, continuous simulation of runoff may be performed, but simpler and less expensive design procedures requiring only IDF or design storm information can be expected to remain in widespread use for the foreseeable future.

Within the United States, information on the magnitudes and frequencies of extreme precipitation is usually obtained from several publications of the National Weather Service (NWS) and its predecessor,

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the Weather Bureau. Those publications are commonly known as TP-40 (Hershfield, 1961), HYDRO-35 (Frederick et al., 1977), and NOAA Atlas 2 (Miller et al., 1973). In all of these publications, rainfall information is presented as a series of maps that illustrate isohyets (i.e. rainfall depth contours) for various storm durations and return periods. Methods are also presented in those publications for interpolation to other durations and/or return periods of interest.

Growing concerns over the ages of the NWS publications, along with recognition of increased gauging record lengths and improved methods of statistical analysis since their publication dates, has motivated a number of newer state-wide and regional studies in the past decade (Durrans and Brown, 2001). This paper concerns one of those studies, performed for the State of Alabama, which stands out in the sense that its products are delivered via the internet. In particular, this paper describes the at-site and regionalized statistical procedures employed to permit estimation of rainfall information for any user-selected site (gauged or ungauged) within the state. The internet-based product delivery approach used for the Alabama Rainfall Atlas served as a model for the similar Precipitation Frequency Data Server currently being implemented by the National Weather Service.

The following section of this paper describes procedures applied to develop 'at-site' estimates of the statistical characteristics of extreme precipitation in Alabama, including a brief discussion of the problems posed by the coarse measurement resolution of many of the raw precipitation data. Additional sections then briefly review a number of the methods that have been suggested for regionalization of statistical information, and then a hybrid approach to the problem is proposed. Comparative evaluations of the performance attributes of the hybrid approach complete the presentation.

2. At-site estimation and databases

As described by Durrans and Brown (2001), the generalized extreme value (GEV) distribution (Jenkinson, 1955) has been selected for modeling of precipitation maxima in Alabama for all gauging sites and storm durations considered. Raw data were available for daily gauging sites, for hourly sites, and for sub-hourly sites. For reasons described shortly, the present paper deals only with those durations ranging from 1 to 48 hours based on hourly data records, and on durations of 1 and 2 days based on daily records.

The GEV distribution has location, scale, and shape parameters, denoted respectively by $\theta_k, k = 1-3$. Estimation of those parameters for each gauging site and storm duration was accomplished independently of all other sites and durations using the direct sample *L*-moment estimators presented by Wang (1996). Correction of the resulting sample *L*-moments for the effects of fixed measurement times using the factors presented by Weiss (1964) was performed, and computation of the parameters θ_k, $k = 1-3$, on the basis of the sample *L*-moments using the expressions presented by Hosking and Wallis (1997) was performed.

Because the GEV distributions fitted to the data for different durations at a fixed gauging site were not always consistent (i.e. they occasionally crossed one another), adjustments were made to the coefficient of *L*-variation (*L-CV*, also known as τ_2) and coefficient of *L*-skewness (*L-CS*, also known as τ_3) so those statistics would vary from site to site, but also so they would be constant for all durations within a given data type (daily, hourly, sub-hourly). At each gauging site and for each type of data, those statistics were computed as the arithmetic averages of the raw statistics computed for each duration. For consistency, raw second *L*-moments, and correspondingly the values of θ_2 and θ_1, were then adjusted for conformance with the adjusted *L-CV*.

Many, if not most, of the raw precipitation data that have been measured and published in the United States since the early 1970 s are reported to the nearest 0.1 in. (2.5 mm). Special methods for coping with the coarseness of that resolution are necessary to avoid serious biases and inflated variances in statistical estimators for short storm durations (Durrans and Pitt, 2004). For the purposes of the Alabama Rainfall Atlas, additional work is necessary to recompute the parameter estimates for sub-hourly (15 and 30-minute) durations and thus results for those durations will not be reported herein.

Databases have been assembled to summarize statistics for daily, hourly, and sub-hourly gauging sites and are relied upon by the internet-based server when responding to user requests. For each daily gauging site, the daily database contains the site identification number, its latitude and longitude, the number of years of record upon which parameters were estimated, and the location, scale, and shape parameters of the GEV distribution for each duration considered for that data type. The hourly and sub-hourly databases were assembled in the same fashion. Storage of information in this manner is very efficient in terms of disk space and means that the information can be manipulated and transported across the internet at high speed. This is believed to be a major consideration given the internet-based delivery method.

3. Review of methods of regionalization

Regionalization is the term given to methods that seek to 'substitute space for time' in hydrologic frequency analysis. The idea is that the limited information available at a gauging site, because of the finite record length and inherent time-sampling variability, can be augmented and enhanced with information available at other hydrologically similar sites.

A number of methods of regionalization have been proposed over the years and have been reviewed and described by Cunnane (1988); Hosking and Wallis (1997). Perhaps the oldest, but still widely applied, method of regionalization is the index-flood procedure of Dalrymple (1960), which assumes that the frequency distributions at all sites within a region are identical except for scale. Dawdy (1961); Benson (1962) observed that flood distributions do not always conform to the index-flood assumption and presented multiple regression procedures that have gained widespread use by the US Geological Survey. Yet another method that has been widely used in the United States is the weighting of a sample coefficient of skewness with a 'map skew' value (IACWD, 1982), though that procedure is coming under increasing criticism (e.g., McCuen, 2001) and may be counterproductive (Landwehr et al., 1978). While much of the literature has dealt with regionalization of flood distributions and terminology such as the 'index flood' has arisen, the methods that have been proposed can be applied with any kind of data.

A difficulty in many methods of regionalization is the need to define homogeneous regions of sites having similar hydrological characteristics. This can be particularly challenging for streamflow (flood, low-flow) distributions because of the influences of drainage basin size, topography and soils, and channel network geometry. Fortunately, methods of regionalization are generally robust in the sense that they yield improvements over at-site analysis even in the presence of moderate distributional heterogeneity (Lettenmaier and Potter, 1985; Lettenmaier et al., 1987; Hosking and Wallis, 1988; Potter and Lettenmaier, 1990). Hosking and Wallis (1993) present several statistics that can be applied to aid the process of defining regions of reasonably homogeneous sites.

A region of hydrologically similar sites, while not necessarily constrained to be a collection of geographically nearby sites, is usually defined in that fashion. Alternative approaches that have been applied include partitioning of sites based on seasonality considerations (Gingras et al., 1994); likelihood-ratio statistics (Wiltshire, 1985); and cluster, factor, and principal components analyses (DeCoursey, 1973; White, 1975; Acreman and Sinclair, 1986; Burn, 1988, 1989; Guttman, 1993). Fiorentino et al. (1987); Gabriele and Arnell (1991) have proposed the use of hierarchical regions where shape parameters are assumed to be constant over large regions, and scale parameters are assumed to be constant over smaller sub-regions. Durrans and Nix (1998); Hamza et al. (2001) have applied simple scaling ideas (Cavadid, 1988) for regionalization of low-flows.

A problem that arises with many methods of regionalization is that discontinuities arise at regional boundaries. A closely related problem is that of assignment of an ungauged site to a region. Wiltshire (1986); Acreman and Wiltshire (1989) proposed a fractional-membership procedure, and Burn (1990) proposed a region of influence procedure to circumvent these difficulties.

4. Regionalization of precipitation estimates for the alabama rainfall atlas

Regionalization of precipitation estimates is generally easier than is the same problem for floods or low-flows, particularly in geographical regions without appreciable topographic relief (Stedinger et al., 1993). In the NWS rainfall atlases TP 40, HYDRO-35, and NOAA Atlas 2, regionalization was implicitly accomplished through the drawing of smooth isohyets of precipitation depth for various storm durations and return periods. Another method that is widely used by the National Weather Service consists of a reciprocal-distance-squared approach, which forms a precipitation estimate at an ungauged site as a weighted average of the estimates at surrounding gauged sites, giving greater weights to sites that are in close proximity to the site of interest (Linsley et al., 1982).

In the Alabama Rainfall Atlas, regionalization of precipitation estimates is accomplished via a hybrid approach consisting of a weighted average of a reciprocal-distance-squared estimator and an index-flood type of estimator like that proposed by Hosking and Wallis (1997). That is, a regionalized estimator, θ_k^{R}, of a parameter $\theta_k, k = 1-3$, for a given duration at a user-specified site of interest is formed as

$$\theta_k^{\mathrm{R}} = \alpha(\theta_{\mathrm{d}})_k^{\mathrm{R}} + (1-\alpha)(\theta_n)_k^{\mathrm{R}}; \qquad 0 \le \alpha \le 1 \tag{1}$$

where $(\theta_{\mathrm{d}})_k^{\mathrm{R}}$ is a reciprocal-distance-squared estimator defined by

$$(\theta_{\mathrm{d}})_k^{\mathrm{R}} = \sum_{i=1}^{N} \frac{\theta_k^{(i)}}{d_i^2} \left(\sum_{i=1}^{N} \frac{1}{d_i^2} \right)^{-1} \tag{2}$$

and $(\theta_n)_k^{\mathrm{R}}$ is an index-flood type of estimator defined by

$$(\theta_n)_k^{\mathrm{R}} = \sum_{i=1}^{N} n_i \theta_k^{(i)} \left(\sum_{i=1}^{N} n_i \right)^{-1} \tag{3}$$

In these expressions, N is the number of gauging sites used to make the estimate, d_i is the distance from the site of interest to the i-th of the N gauging sites, n_i is the record length for the i-th of the N gauging sites, and $\theta_k^{(i)}$ is the estimate of the parameter θ_k at the ith of the N gauging sites. Note that Eq. (2) gives larger weights to nearby gauging sites than to those that are farther away, whereas Eq. (3) gives larger weights to sites with long record lengths. Thus, provided that α is neither zero nor unity, the estimator given by Eq. (1) addresses both closeness and record length. The hybrid estimator, Eq. (1), makes use of 'redundant' estimators, Eqs. (2) and (3), and thus may be expected to be superior to both of them (Gelb, 1974).

The reader should note that when $\alpha = 1$, Eq. (1) is a 'spatial interpolator' in the sense that a regionalized estimate of a statistic at a gauged site is simply equal to the corresponding statistic at that site, i.e. $\theta_k^{\mathrm{R}} = \theta_k^{(i)}$. This happens of course because the weight assigned to the statistic at that site is infinite; no information from other gauging sites is brought to bear on the regionalized estimate and the whole intent of regionalization is defeated. Provided that α is constrained to be less than unity, the estimator given by Eq. (1) is a 'spatial smoother,' as desired.

Eq. (1) can be easily generalized to permit combination of more than two regionalized (redundant) estimators of a statistic. By inclusion of additional weighting factors one could, for example, incorporate the effects of land surface elevation or slope aspect as well as site nearness and record length into the final weighted estimate. Any resulting estimator would be assured of being a 'spatial smoother' as long as at least one of the factors (such as record length) itself resulted in a smoothing rather than interpolating effect. Of course, the weights would also have to sum to unity for the resulting estimator to be unbiased.

5. Performance of the hybrid estimator

Testing of the performance of Eq. (1), and developing estimates of the weighting parameter α, has been accomplished using a cross-validation method. The procedure involved estimation of a known parameter at a gauging site using only the information available at all other gauging sites, and then comparing the known and estimated values. By performing this sequentially for all sites one can develop estimates of the bias and mean squared error of the hybrid estimator for various selections of the weighting parameter α. This can then serve as a basis for selecting an optimal estimate of that parameter.

Table 1 presents a summary of optimal values of α for each parameter θ_k and for each storm duration. The values of α shown are those that are optimal in terms of the bias of the hybrid estimator. Also shown in the table are the biases and associated mean square errors of the hybrid estimator. It is seen that the biases are zero, and that the optimum value of α tends to decrease slightly as the storm duration increases (except for the parameter θ_3, where the optimal α is

Table 1
Optimal values of α for minimization of bias

Data type	Duration	Optimal α	Bias	Associated MSE
Location parameter, θ_1				
Hourly	1 h	0.9	0.00	0.17
	2 hr	0.9	0.00	0.20
	4 hr	1.0	0.00	0.22
	6 hr	0.9	0.00	0.26
	12 hr	0.8	0.00	0.33
	24 hr	0.7	0.00	0.42
	48 hr	0.7	0.00	0.46
Daily	1 day	0.8	0.00	0.33
	2 days	0.7	0.00	0.38
Scale parameter, θ_2				
Hourly	1 h	0.8	0.00	0.09
	2 h	0.8	0.00	0.11
	4 h	1.0	0.00	0.14
	6 h	0.9	0.00	0.16
	12 h	0.7	0.00	0.19
	24 h	0.6	0.00	0.23
	48 h	0.6	0.00	0.26
Daily	1 day	0.9	0.00	0.23
	2 days	0.8	0.00	0.26
Shape parameter, θ_3				
Hourly	1 h	0.25	0.00	0.14
	2 h	0.25	0.00	0.14
	4 h	0.25	0.00	0.14
	6 hr	0.25	0.00	0.14
	12 h	0.25	0.00	0.14
	24 h	0.25	0.00	0.14
	48 h	0.25	0.00	0.14
Daily	1 day	1.0	0.00	0.16
	2 days	1.0	0.00	0.16

Table 2
Optimal values of α for minimization of mean squared error

Data type	Duration	Optimal α	MSE	Associated Bias
Location parameter, θ_1				
Hourly	1 h	1.0	0.17	0.00
	2 h	1.0	0.20	0.00
	4 h	1.0	0.22	0.00
	6 h	1.0	0.26	0.00
	12 h	1.0	0.32	0.01
	24 h	1.0	0.40	0.01
	48 h	1.0	0.43	0.01
Daily	1 day	1.0	0.32	−0.01
	2 days	1.0	0.36	−0.01
Scale parameter, θ_2				
Hourly	1 h	0.7	0.09	0.00
	2 h	0.8	0.11	0.00
	4 h	0.9	0.14	0.00
	6 h	0.9	0.16	0.00
	12 h	0.8	0.19	0.00
	24 h	0.8	0.22	0.00
	48 h	0.8	0.25	0.00
Daily	1 day	1.0	0.23	0.00
	2 days	1.0	0.25	0.00
Shape parameter, θ_3				
Hourly	1 h	0.55	0.14	0.00
	2 h	0.55	0.14	0.00
	4 h	0.55	0.14	0.00
	6 h	0.55	0.14	0.00
	12 h	0.55	0.14	0.00
	24 h	0.55	0.14	0.00
	48 h	0.55	0.14	0.00
Daily	1 day	0.4	0.16	−0.01
	2 days	0.4	0.16	−0.01

independent of the duration). This seems to make physical sense, as the larger values of α tend to be associated with the shorter durations, where the annual rainfall maxima are generally associated with localized convective activity, and hence where the distance-based estimator would be expected to be a good performer. Mean squared errors shown in Table 1 generally increase with the storm duration and tend to be higher for the location and scale parameters than for the shape parameter.

Table 2 is similar to Table 1, but shows values of α that are optimal in terms of the mean squared error of the hybrid estimator. Also shown in Table 2 are the mean squared errors and associated biases of the hybrid estimator. A comparison between Tables 1 and 2 shows that while the optimal value of α is different when bias as opposed to when mean squared error is the criterion, little is lost by preferring one to the other. That is, biases associated with optimization of mean squared error are not much different than the optimal biases. Similarly, mean squared errors associated with optimization of bias are not much different than the optimal mean squared errors.

Table 3 presents a comparison of the optimal mean squared error of the hybrid estimator to the mean squared errors associated with each of its components (i.e. the reciprocal-distance-squared and index-flood estimators). It may be observed from the table that the mean squared error associated with the hybrid estimator is smaller than or equal to the errors associated with each of the component estimators

Table 3
Comparison of optimal mean squared errors of hybrid estimator to the mean squared errors of the index-flood (MSE_n) and reciprocal-distance-squared (MSE_d) estimators

Data type	Duration	Optimal MSE	MSE_n	MSE_d
Location parameter, θ_1				
Hourly	1 h	0.17	0.23	0.17
	2 h	0.20	0.30	0.20
	4 h	0.22	0.37	0.22
	6 h	0.26	0.42	0.26
	12 h	0.32	0.49	0.32
	24 h	0.40	0.56	0.40
	48 h	0.43	0.66	0.43
Daily	1 day	0.32	0.43	0.32
	2 days	0.36	0.47	0.36
Scale parameter, θ_2				
Hourly	1 h	0.09	0.10	0.09
	2 h	0.11	0.13	0.11
	4 h	0.14	0.17	0.14
	6 h	0.16	0.19	0.16
	12 h	0.19	0.23	0.20
	24 h	0.22	0.27	0.23
	48 h	0.25	0.31	0.26
Daily	1 day	0.23	0.29	0.23
	2 days	0.25	0.32	0.25
Shape parameter, θ_3				
Hourly	1 h	0.14	0.14	0.14
	2 h	0.14	0.14	0.14
	4 h	0.14	0.14	0.14
	6 h	0.14	0.14	0.14
	12 h	0.14	0.14	0.14
	24 h	0.14	0.14	0.14
	48 h	0.14	0.14	0.14
Daily	1 day	0.16	0.16	0.16
	2 days	0.16	0.16	0.16

and hence that the hybrid estimator is superior to both of them. As pointed out earlier, this should be expected as a consequence of the use of redundant information with a hybrid estimator.

An additional and interesting observation from Table 3 is that the mean squared error associated with the reciprocal-distance-squared estimator is nearly always less than the mean squared error associated with the index-flood estimator. This suggests that further investigation may be in order, particularly in view of the widespread use of the index-flood approach. This result, combined with the earlier observation that little is lost in terms of bias when α is chosen on the basis of mean squared error, has led to the adoption of the optimal α values listed in Table 2. While some of those α values are equal to one, implying that only distance (and not record length) is important, and hence that the hybrid estimator is a spatial interpolator rather than a smoother, values of $\alpha = 1$ do not occur for all three distributional parameters simultaneously for any storm duration. Thus, even though one (or sometimes two) of the parameters of the distribution may be interpolated, the remaining parameter(s) is (are) smoothed and hence the distribution as a whole is smoothed.

6. Conclusions

A hybrid estimator for regionalization of precipitation frequency information has been presented and evaluated in this paper. The hybrid estimator makes optimal use of the redundant information provided by reciprocal-distance-squared and index-flood estimators and has performance attributes superior to those exhibited by the two component estimators. The estimator is easily applied and is operationally fast. It is therefore quite suitable for implementation in an internet-based product delivery platform.

The potential advantages offered by the use of redundant information suggest that future research should consider the possibility of incorporating additional information, such as site elevation and land surface slope aspect, into the hybrid regionalization method. Additional work also needs to be accomplished to extend the methods presented here to parameters for sub-hourly storm durations, which will require further inquiry and analysis of the problems posed by coarse data resolutions.

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